

KAN-Based Adaptive Parameter Control for Multi-User Satellite FSO/QKD Systems

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Abstract—For multi-user satellite free-space-optical quantum key distribution with subcarrier-intensity-modulation BPSK and dual-threshold detection, we optimize modulation depth, dual-threshold coefficients and the inter-user exclusion ratio under unauthorized-receiver and beam-splitting security, a nonconvex problem coupling all N pairs. A two-level decomposition with an exact outer/inner split is the oracle for a learned controller, and we compare Kolmogorov–Arnold Networks (KAN), regularized multi-layer perceptrons (MLP) and a linear map. Over an SGP4-propagated Walker shell at Hanoi, adaptation yields $3.6\times$ the secure pair-time on the reported epoch, $6.2\times$ pooled over 10; on a sparser shell it yields $8.4\times$ pooled, raising a static ceiling of 1 secure pair-step to a floor of 2: survival rather than operation. Three of four outputs are constant because the modulation depth saturates, so accuracy is reported on the one with signal. KAN and the MLP are indistinguishable on 18 of 20 seeds; the MLP degenerates to the linear fit on 2 and the KAN on none, which this sample size does not establish ($p = 0.29$). Retention saturates for all three and is not ranked. What separates them is parameter count and the closed-form rule extractable from the KAN, post-extraction $R^2 \approx 0.97$ in the single-link setting, at a latency cost.

Reproducibility. All code, bundled TLE data, and run scripts that reproduce every figure and table in this paper are available at <https://github.com/haodpsut/kan-satellite-fso-qkd>; the head-line runs reproduce in under one hour on a single CPU plus an RTX-class GPU for the KAN training phase.

Index Terms—Quantum key distribution, free-space optical communications, satellite networks, Kolmogorov–Arnold Networks, multi-user QKD, adaptive control.

I. INTRODUCTION

SATELLITE-based free-space optical (FSO) channels are the most credible medium for intercontinental quantum key distribution (QKD): they avoid the exponential photon-loss bound of optical fiber and exploit the near-vacuum of space for low excess noise [1], [2], [3], [4]. The 2017 demonstration of decoy-state BB84 between the Micius satellite and ground stations in China [1], the simultaneous entanglement distribution at 1,200 km baseline [5], and the relayed intercontinental key exchange of [6] together establish satellite QKD as a near-term technology rather than a research curiosity. Recent progress is summarized in [3], [4]. Two reviews frame the present work in particular: [7] surveys how device imperfections, rather than protocol design, set the security of practical QKD, which is

the regime our parameter-control problem lives in; and [8] reviews the Micius programme, still the only satellite platform on which these protocols have been demonstrated end to end, and the source of the link budgets that make a multi-user relay architecture worth optimizing at all.

Among the QKD families adapted to FSO, the subcarrier-intensity-modulation BPSK with dual-threshold direct detection (SIM-BPSK/DT-DD) scheme proposed by Vu, Pham, Dang, Le, and Pham [9], [10], [11] is attractive because it relies on entirely classical components (an EDFA-amplified GEO source, PIN photodetectors, and dual-threshold post-processing), yet is reported in [9], [10], [11] to admit a positive secret-key rate against unauthorized-receiver attacks (URA) and beam-splitting attacks (BSA) [12], [13], provided that the modulation depth μ , the DT scale coefficients β_A, β_i , and the multi-user fading-randomness exclusion ratio χ are appropriately chosen. We state the scope of that guarantee precisely, because the present work inherits it rather than re-deriving it: the secret-key rate used throughout this paper is the asymptotic Csiszár–Körner fraction $\max\{0, H_2(P_e) - H_2(\text{QBER})\}$ per sifted bit, evaluated against the two attack models above. It carries no finite-key correction, no composable-security statement, and no claim of optimality against general coherent attacks; it is a sufficient-condition surrogate that we optimize, not a security proof that we establish. SIM-BPSK is well established for classical FSO links [14], [15], and DT-DD inherits its noise-budget structure from the standard outage-capacity analysis [16], [17].

The three papers in the lineage establish, in increasing generality, the relay-assisted single-link case [9], the entanglement-based two-user case [10] (which adapts the BBM92 protocol [18] to the DT/DD measurement basis), and the GEO source serving N ground users via dual LEO relays with a closed-form inclusion-exclusion term [11]. In all three, the parameters (μ, β, χ) are picked *statically*: a single operating point is selected by reading the analytic performance plots at a representative geometry, and that point is then held throughout operation.

We show in this paper that for the realistic case of a LEO relay traversing its pass while serving an asymmetric cluster of ground users, the static-parameter assumption is the dominant performance bottleneck. Peak elevation, slant range, log-amplitude turbulence variance [19], [20], and the URA leakage geometry vary by orders of magnitude across a single 5 to 10 minute pass, and the optimal $\mathbf{p}(t) = (\mu, \beta_A, \{\beta_i\}, \chi)$ drifts continuously with the time-varying state $\mathbf{s}(t)$. Under realistic constellations propagated from real Two-Line Elements (TLE) via the SGP4 propagator [21], [22] on Starlink-density shells

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[23], [24], we further show that *no* single static $\mathbf{p}$ produces any secure pair-step over a typical 90-minute observation window, so adaptation is a precondition for the system to operate at all rather than a marginal improvement.

The modulation-signalling branch of QKD that this paper belongs to has moved on since the three papers above, and in directions that sharpen what is and is not new here. Subcarrier-wave QKD has been analysed numerically for free-space ship-to-port links [25] and routed through a deployed metropolitan transport network [26], which establishes the signalling format outside the laboratory but in both cases for a *single* link rather than a shared relay. On the transmitter side, direct intensity and phase modulation has been demonstrated as a QKD source [27], which is the hardware assumption our modulation depth parameterizes. The free-space continuous-variable line is surveyed in [28]; it shares our reliance on classical coherent components but replaces the dual-threshold decision with homodyne detection, so the parameter that dominates our problem, a detection threshold, has no counterpart there. What none of these address is the coupling we study: a single global modulation depth that every user in a cluster must share.

Learning-based parameter control for QKD is itself an active line, and the closest neighbour to this work is [29], which couples a temporal convolutional forecaster with a proximal-policy-optimization agent to retune decoy intensities online for BB84, E91 and COW while keeping a composable-security constraint. The contrast is instructive in three ways. First, the physical layer differs: that work tunes photon-counting discrete-variable protocols, whereas the parameters here are a modulation depth and a pair of detection thresholds in an intensity-modulated direct-detection link, so the two parameter spaces do not overlap and a numerical head-to-head comparison would compare protocols rather than controllers. Second, that work is single-link; the coupling that makes the present problem hard, a single global modulation depth shared by every Alice-Bob pair in a cluster, does not arise there. Third, it learns online by reinforcement while we learn offline by supervision from a decomposed oracle, which buys determinism and an inspectable target at the cost of needing that oracle. A reinforcement-learning controller for the multi-user FSO setting is a worthwhile comparison that we leave to future work; building a fair one is a study in itself, and a weak re-implementation would be worse evidence than none.

A. Contributions

This paper makes the following contributions:

- 1) **Full multi-user joint problem.** We restate the GEO/multi-LEO/ N -user system of [11] together with the pointing-error/Gaussian-beam model of [9] (dropped in [11]) and derive the corresponding joint problem $(\mathbf{P})$ over $\mathbf{p} = (\mu, \beta_A, \{\beta_i\}, \chi)$ with URA, BSA, and an inter-user secrecy constraint. The problem is nonconvex with three sources of coupling: μ shared globally, χ entering every pair through a product field $\phi_i = \prod_{j \neq i} (1 - P_c^j)$, and β_A shared between Alice and every Bob. The inclusion-exclusion term of [11, Eq. 22] is generalized to asymmetric clusters in closed form.

- 2) **Principled two-level decomposition.** We expose the variable roles to obtain a two-level decomposition whose outer/inner split is exact (a relaxation-free reformulation of $(\mathbf{P})$): an outer two-dimensional search over (μ, χ) wraps an inner mean-field block-coordinate solver over the thresholds. The symmetric-cluster fixed point provably recovers [11, Eq. 24], and the decomposition matches a brute-force joint grid optimum on a two-user cluster to four decimals.
- 3) **Two-stage controller with KAN.** The decomposed oracle generates supervised pairs $(s(t) \rightarrow p^*(t))$ for a two-stage controller: a global head consumes cluster-aggregate features and predicts (μ, χ, β_A) ; a per-user shared head consumes $(s_i, \beta_A, \mu, \chi)$ and predicts β_i . We compare a Kolmogorov–Arnold Network (KAN) [30], [31] against a regularized multi-layer perceptron and a linear residual baseline on a five-axis trade-off (regression accuracy, closed-loop secret-key retention, parameter count, inference latency, and interpretability). Deep learning for the classical physical layer [32], [33], [34] and reinforcement-learning control [35] are well studied; KAN is attractive here because its spline activations admit symbolic extraction [31], [36], which yields a closed-form rule for β^* .
- 4) **TLE-driven realistic-orbit validation.** An SGP4 path mirrors the analytic-orbit API so every script accepts `-tle` as a drop-in switch, with a synthetic Walker shell reproducing Starlink coverage statistics offline and a Celestrak bundle loadable via `-tle PATH`. Over a Hanoi ground station and 10 TLE epochs, the best static choice on the sparse Walker shell never exceeds 1 secure pair-step per epoch, and reaches zero on 5 of them, while adaptation never falls below 2.
- 5) **Honest multi-axis evaluation.** We report results with explicit mean $\pm$ std over five seeds, fair (regularized) MLP baselines, and a model-agnostic β safety-margin recipe that lifts feasibility by 1.1–1.5 $\times$ for every controller. KAN does not dominate every axis; its value is parameter efficiency plus an interpretable closed-form rule for β with symbolic $R^2 \approx 0.97$. We disclose where the symbolic extraction fails (the μ^* saturation regime), where the controllers transfer (single-link) and where they do not (multi-user cluster, the residual-learning rescue), and how the design generalizes to real-orbit operation.

II. SYSTEM MODEL

A. Architecture

Table I collects the symbols used in this section and in Sections III–IV.

A geostationary source *Charlie* (C) at altitude $H_C=35\,793$ km broadcasts a SIM-BPSK-modulated continuous-wave laser of mean power $P=32$ dBm and wavelength $\lambda=1550$ nm. The signal is amplify-and-forward relayed by two low-Earth-orbit satellites ($H_L=550$ km, Starlink shell) acting as optical amplifiers (EDFA gain $G_a=40$ dB). The relayed beam illuminates a ground

TABLE I
NOTATION USED IN SECTIONS II–IV.

Symbol	Meaning
<i>Decision variables</i>	
μ	modulation depth of the SIM-BPSK source, $\mu \in (0, 1)$
β_A	dual-threshold scale coefficient at Alice
β_i	dual-threshold scale coefficient at user B_i
χ	inter-user fading-randomness exclusion ratio
$\mathbf{p}$	parameter vector $(\mu, \chi, \beta_A, \{\beta_i\})$
<i>Geometry and channel</i>	
N	number of ground users in the cluster
L	link distance
θ_U	zenith angle of link U
D, w_0	receive aperture diameter, transmit beam waist
h_{e2e}^U	end-to-end channel coefficient of link U , $h_g^U h_l^U h_a^U$
$\bar{h}^U$	deterministic (geometric and atmospheric) part of h_{e2e}^U
h_a	log-normal atmospheric-turbulence fading term
σ_X	log-amplitude standard deviation of the turbulence
$C_n^2(h)$	refractive-index structure parameter, Hufnagel–Valley
γ^U	mean SNR amplitude of link U , $\gamma^U \propto \mu$
w_{eq}	equivalent beam width at the receiver
d_E	Eve's standoff distance from the legitimate receiver
<i>Detection and key rate</i>	
R_e, P, G_a	photodetector responsivity, source power, EDFA gain
R_b	system bit rate
P_{sift}	sifting probability after dual-threshold detection
QBER	quantum bit error rate on the sifted bits
P_{err}	Eve's bit-error probability (URA, optimal threshold $d_E=0$)
H_2	binary entropy function
σ_i	per-pair secret fraction, $\max\{0, I(A; B_i) - \max_E I(\cdot; E)\}$
R_s^i, R_f	per-pair sifted-key rate; cluster key rate, the objective of (P)
<i>Solver and controller</i>	
$\mathcal{M}_\mu, \mathcal{M}_\chi$	outer grids over μ and χ
ϕ_i	mean-field term seen by user i in the inner iteration
$\epsilon_\phi, \epsilon_R$	inner-loop tolerances on the field and on the rate
$\mathbf{s}$	controller input (cluster state)
$\mathbf{p}^*$	oracle-optimal parameter vector for a state
δ	additive safety margin applied to a predicted β

cluster consisting of a single server *Alice* (A) and N users $B_1, \dots, B_N$; an entanglement-based BBM92-style key is then established *per pair* (A, B_i) using only classical optical components and dual-threshold post-processing. An eavesdropper *Eve* sits at radial offset d_E from each legitimate receiver. The architecture is depicted in Fig. 1.

B. Channel Model

The end-to-end channel coefficient for any link $C \rightarrow U$ ($U \in \{A, B_i, E\}$) factors as

$$h_{e2e}^U = h_g^U h_l^U h_a^U, \quad (1)$$

where the geometric, attenuation, and turbulence components are obtained as follows.

1) *Gaussian-beam spreading*: At link distance L a beam of waist $w_0 = \lambda/(\pi\theta)$ (divergence $\theta = 2.44\lambda/D$) has radius $w_L = w_0[1 + (\lambda L/(\pi w_0^2))^2]^{1/2}$ [19]. The fraction collected by a receiver aperture of radius a at radial offset r from the

footprint center is, after the small-aperture approximation of [16],

$$h_g(r) \approx A_0 \exp\left(-\frac{2r^2}{w_{L,eq}^2}\right), \quad A_0 = [\text{erf}(v)]^2, \quad (2)$$

$$v = \frac{\sqrt{\pi} a}{\sqrt{2} w_L}, \quad w_{L,eq}^2 = w_L^2 \frac{\sqrt{\pi} \text{erf}(v)}{2v e^{-v^2}}. \quad (3)$$

For Eve at $r = d_E$ the geometric leakage ratio $h_g^E/h_g^U = \exp(-2d_E^2/w_{L,eq}^2)$ is close to unity for realistic d_E (tens of metres) versus footprint widths of hundreds of metres; security must therefore come from small μ rather than from geometry alone.

2) *Atmospheric attenuation*: Beer–Lambert with visibility-dependent attenuation

$$h_l = e^{-\sigma L_{h-U}}, \quad \sigma(\lambda) = \frac{3.912}{V} \left(\frac{\lambda[\text{nm}]}{550}\right)^{-q(V)}, \quad (4)$$

where $L_{h-U} = (H_h - H_U)/\cos\theta_U$, $H_h=20$ km, and the size-distribution coefficient is the standard piecewise law $q(V) = 1.6$ for $V > 50$ km, $q(V) = 1.3$ for $6 < V \leq 50$ km, and $q(V) = 0.585 V^{1/3}$ for $V \leq 6$ km.

3) *Log-normal turbulence*: For elevations $\geq 30^\circ$ (zenith $\leq 60^\circ$) the turbulence is weak and h_a is log-normal with unit mean [19], [20]: $f_{h_a}(h_a) = \frac{1}{\sqrt{8\pi h_a \sigma_X}} \exp(-(\ln h_a - 2\mu_X)^2/(8\sigma_X^2))$ with $\mu_X = -\sigma_X^2$ enforcing $\mathbb{E}[h_a] = 1$. The log-amplitude variance is the (quarter-)Rytov variance

$$\sigma_X^2 = 0.56 k^{7/6} \sec^{11/6}(\theta_U) \int_{H_U}^{H_C} C_n^2(h) (h - H_U)^{5/6} dh, \quad (5)$$

with the Hufnagel–Valley profile $C_n^2(h) = 0.00594(w/27)^2(10^{-5}h)^{10}e^{-h/1000} + 2.7 \times 10^{-16}e^{-h/1500} + C_n^2(0)e^{-h/100}$ and $k = 2\pi/\lambda$.

C. Detection in Reduced Form

Define the deterministic mean peak signal current

$$s^U = \frac{1}{4}\mu R_e P G_a \bar{h}^U, \quad \bar{h}^U = h_g^U h_l^U, \quad (6)$$

and the normalized mean SNR amplitude $\gamma^U \triangleq s^U/N^U$, where $(N^U)^2$ is the AWGN variance budgeted in [11, Eq. 4] (shot, background, ASE, thermal). For the DT scale coefficient β^U , the conditional probabilities collapse to two log-normal expectations

$$P_{\text{corr}}^U = \mathbb{E}_{h_a}[Q(\beta^U + \gamma^U(1 - h_a))], \quad (7)$$

$$P_{\text{err}}^U = \mathbb{E}_{h_a}[Q(\beta^U + \gamma^U(1 + h_a))], \quad (8)$$

both evaluated by Gauss–Hermite quadrature with $h_a(x_j) = \exp(2\sqrt{2}\sigma_X x_j - 2\sigma_X^2)$ [11, Eq. 30]; the order- n nodes and weights are tabulated in [37, Tbl. 25.10], and $n=20$ converge to numerical precision for the values of σ_X encountered along a LEO pass. The per-link sift probability and QBER follow as $P_{\text{sift}}^U = P_{\text{corr}}^U + P_{\text{err}}^U$ and $\text{QBER}^U = P_{\text{err}}^U/P_{\text{sift}}^U$.

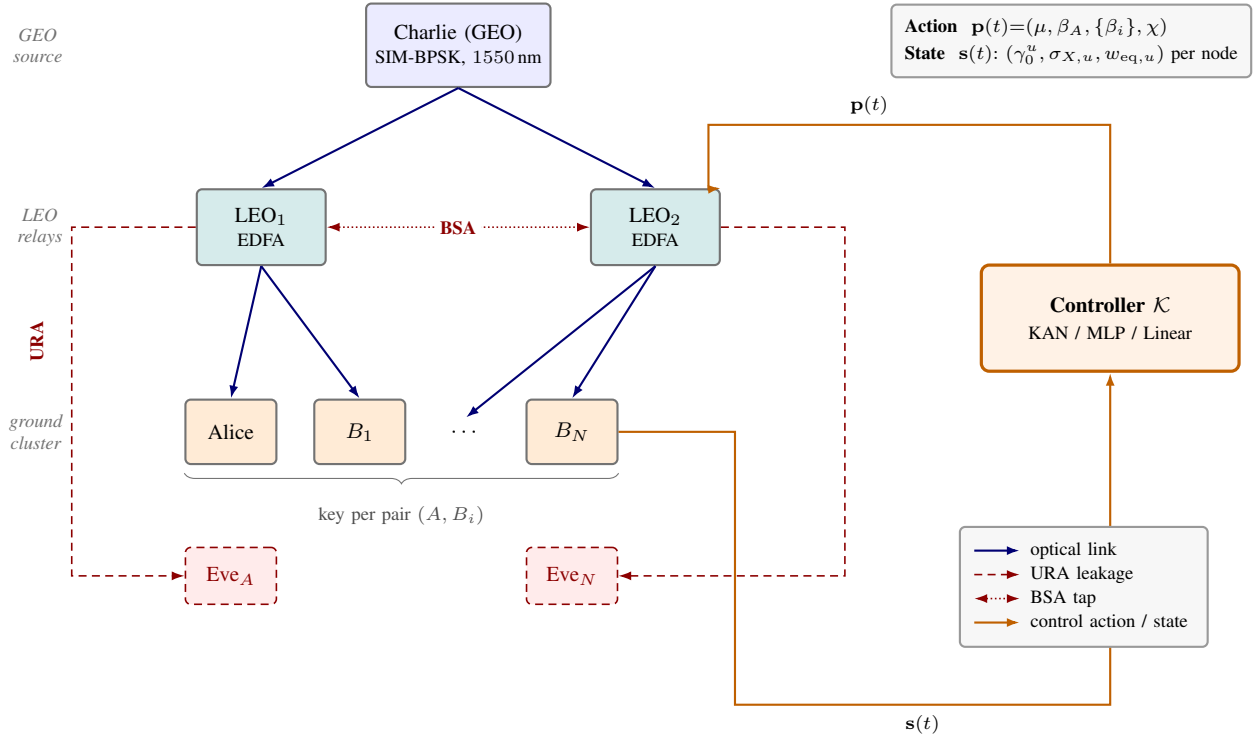


Fig. 1. System architecture. A GEO source Charlie broadcasts a SIM-BPSK signal through two LEO EDFA relays to a ground cluster (Alice + N Bobs); the secret key is established per pair (A, B_i) . An eavesdropper Eve at offset d_E from each receiver mounts an unauthorized-receiver attack; a beam-splitter at the LEO relay mounts a beam-splitting attack. The proposed controller selects $\mathbf{p}(t) = (\mu, \beta_A, \{\beta_i\}, \chi)$ from the time-varying cluster state $\mathbf{s}(t)$.

D. Pairwise Sift, Exclusion, and Secret Key Rate

For the pair (A, B_i) the raw sift is $P^{AB_i} = P_{\text{sift}}^A P_{\text{sift}}^i$ and the QBER is

$$\text{QBER}^{AB_i} = \frac{P_{\text{corr}}^A P_{\text{err}}^i + P_{\text{err}}^A P_{\text{corr}}^i}{P_{\text{sift}}^A P_{\text{sift}}^i}. \quad (9)$$

The proposed multiple-access scheme retains only bits not coincidentally accepted by other pairs. Assuming statistically independent acceptance across users (independent per-link fading), inclusion–exclusion over the asymmetric cluster yields the closed-form excluded sift

$$P_{\text{excl}}^{AB_i} = P_{\text{corr}}^A P_{\text{corr}}^i \left[1 - \prod_{j \neq i} (1 - P_{\text{corr}}^j) \right], \quad (10)$$

which generalizes the symmetric expression of [11, Eq. 22]. We verify in the unit tests that (10) reproduces a direct subset-enumeration inclusion–exclusion sum to 10^{-12} for $N \in \{1, 2, 4, 6\}$. With $\chi \in [0, 1]$ controlling the strength of the exclusion,

$$P_{\chi}^{AB_i} = P_{\text{excl}}^{AB_i} - \chi P_{\text{excl}}^{AB_i}, \quad R_s^i = P_{\chi}^{AB_i} R_b. \quad (11)$$

The per-pair secret fraction (binary-erasure wiretap, Eve₁ near Alice, Eve₂ near B_i) is

$$\sigma_i = \max(0, I(A; B_i) - \max(I(A; E_1), I(B_i; E_2))), \quad (12)$$

with $I(A; B_i) = 1 - H_2(\text{QBER}^{AB_i})$ and $I(U; E) = 1 - H_2(P_{\text{err}}^E)$. The cluster key rate, the objective of $(\mathbf{P})$, is

$$R_f(\mathbf{p}; \mathbf{s}, t) = \sum_{i=1}^N R_s^i \sigma_i. \quad (13)$$

III. JOINT OPTIMIZATION PROBLEM

A. Decision Variables and Couplings

At each time instant t of a LEO pass, the cluster state

$$\mathbf{s}(t) = (s_A(t), s_1(t), \dots, s_N(t), d_E) \quad (14)$$

is computed from the time-varying geometry via Sec. II-B; each $s_u = (\gamma_0^u, \sigma_{X,u}, w_{\text{eq},u})$ captures the per-node SNR scale, log-amplitude variance, and beam width. The controller selects

$$\mathbf{p}(t) = (\mu, \beta_A, \{\beta_i\}_{i=1}^N, \chi) \quad (15)$$

with $\mu \in (0, 1)$, $\beta \in [\beta_{\text{lo}}, \beta_{\text{hi}}]$, and $\chi \in [0, 1]$. Three sources of coupling distinguish $(\mathbf{P})$ from a per-link problem:

- *Global* μ . The modulation depth feeds into every $\gamma^u = \gamma_0^u \mu$ and is the URA security lever ($P_{\text{err}}^E \rightarrow 0.5$ as $\mu \rightarrow 0$).
- *Shared* β_A . Alice's threshold enters every pair through P_{corr}^A and P_{err}^A in (9); choosing β_A to favor one Bob hurts the others.
- *Field* χ . The exclusion term (10) couples every pair through the field $\phi_i = \prod_{j \neq i} (1 - P_{\text{corr}}^j)$; binding χ to enforce inter-user secrecy hurts the asymptotically weakest user the most.

B. Threats and Constraints

Two threats are modeled.

1) *Unauthorized-Receiver Attack (URA)*: An eavesdropper near each legitimate node uses the optimal decision threshold $\beta^E = 0$ (she cannot know the legitimate β). Her error probability with her own ratio $\gamma^E = s^E/N^E$ (where $s^E \propto h_g^E$ via (2)) is

$$P_{\text{err}}^E = \mathbb{E}_{h_a}[Q(\gamma^E h_a)]. \quad (16)$$

As $\mu \rightarrow 0$ we have $\gamma^E \rightarrow 0$ and $P_{\text{err}}^E \rightarrow Q(0) = 1/2$. The security target is $P_{\text{err}}^E > p_e^{\min} = 0.1$ [11, Sec. V-B]; the same wiretap idea underlies the classical secrecy-rate bound of [38], generalized to the practical-QKD setting in [12], [39].

2) *Beam-Splitting Attack (BSA)*: An attacker installs a passive beam splitter at the LEO relay drawing fraction η of the optical power; this is the dominant side-channel threat against weak-coherent-state QKD [13], [12]. Her presence is detectable as a sift-probability deviation

$$m_{\text{BSA}}(\beta) = 1 - \frac{P_{\text{sift}}(\gamma(1 - \eta_{\text{leak}}))}{P_{\text{sift}}(\gamma)}, \quad (17)$$

which is monotonically increasing in β . We require $m_{\text{BSA}}(\beta_A), m_{\text{BSA}}(\beta_i) \geq m^{\min} = 0.5\%$ at the assumed split $\eta_{\text{leak}} = 1.5\%$. This pair of numbers sets the *detection sensitivity* of the design, and it is sharper than it looks. Sweeping η_{leak} over the oracle solutions of all 250 clusters, the fraction of pairs for which detection can be certified falls from 100% at $\eta_{\text{leak}} \geq 1.4\%$ to 73% at 1.25% and to zero at 1.15%: the whole transition occupies about 0.25 percentage points. The operating point therefore guarantees detection of a splitter drawing 1.4% or more of the optical power; against a stealthier splitter the constraint is simply not met and the protocol emits no key, which is the conservative outcome but also a real limit on what this design certifies. This imposes a *lower* bound on every β , complementary to the operational *upper* bound from P_{sift} .

3) *Inter-User Secrecy*: The unkept fraction $(1 - \chi)P_{\text{excl}}^{AB_i}/P_{\chi}^{AB_i}$ leaks to other users; we cap it at $\tau = 0.05$. The specific value is not load-bearing: sweeping τ over the oracle solutions of all 250 clusters leaves the feasible-pair count and the aggregate key rate *unchanged* for every $\tau \geq 0.05$, and the constraint only becomes active below $\tau \approx 0.04$, where the feasible count falls from 766 to 608. We therefore report τ as an inactive guard in the studied regime rather than as a tuned parameter.

C. Problem Statement

The full joint problem is

$$\begin{aligned} & \max_{\mu, \beta_A, \{\beta_i\}, \chi} R_f(\mathbf{p}; \mathbf{s}, t) \\ \text{s.t. } & \sigma_i > 0, P_{\chi}^{AB_i} > p_s^{\min}, \text{QBER}^{AB_i} \leq q, \\ \text{(P): } & P_{\text{err}}^E(A), P_{\text{err}}^E(B_i) > p_e^{\min}, \\ & m_{\text{BSA}}(\beta_A), m_{\text{BSA}}(\beta_i) \geq m^{\min}, \\ & (1 - \chi)P_{\text{excl}}^{AB_i}/P_{\chi}^{AB_i} \leq \tau, \\ & \mu \in (0, 1), \beta \in [\beta_{\text{lo}}, \beta_{\text{hi}}], \chi \in [0, 1], \end{aligned} \quad (18)$$

where the first four constraint lines hold for every user $i = 1, \dots, N$.

D. Why (P) Is Hard

Three properties make (P) unsuitable for closed-form treatment:

- 1) Each constraint depends on Gauss–Hermite integrals (7)–(8), (16) of $Q(\cdot)$ over h_a ; no analytic stationarity conditions in β .
- 2) The map $\mathbf{p} \mapsto R_f$ couples $N + 1$ thresholds through the exclusion field ϕ_i and through the shared β_A ; coordinate-descent over thresholds without explicit field tracking does not converge.
- 3) The feasibility set is a thin, multi-directional boundary in $(\mu, \chi, \beta_A, \beta_i)$ space; small parameter errors flip a pair infeasible and forfeit its key.

We exploit the variable roles in Sec. IV to obtain an exact two-level decomposition, the oracle for the learned controller of Sec. V.

IV. DECOMPOSITION AND SOLVER

A. Variable Roles and Two-Level Structure

The decision variables of (P) naturally split into a *global/coupling block* (μ, χ) and a *link-local block* $(\beta_A, \{\beta_i\})$. Fixing the global block, the inner problem

$$\begin{aligned} \text{(I): } V(\mu, \chi) &= \max_{\beta_A, \{\beta_i\}} \sum_{i=1}^N R_s^i \sigma_i \\ &\text{s.t. per-pair constraints of (18),} \end{aligned} \quad (19)$$

returns a value function; the outer problem

$$\text{(O): } \max_{\mu \in (0, 1), \chi \in [0, 1]} V(\mu, \chi) \quad (20)$$

is then a two-dimensional search amenable to a coarse grid plus local refinement. The outer/inner *split* is a lossless reformulation of (P), and it is worth stating why in one line, since the two halves of the solver have very different status. Writing the feasible set as $\mathcal{F} = \{(\mu, \chi, \beta)\}$ and the objective as $R_f(\mu, \chi, \beta)$, partial maximization gives $\max_{\mathcal{F}} R_f = \max_{(\mu, \chi)} [\max_{\beta \in \mathcal{F}(\mu, \chi)} R_f]$ whenever the inner maximum exists, which it does because β ranges over a compact box and R_f is continuous on it. No relaxation, no bound, and no assumption beyond compactness is involved: the identity is exact. What the identity does *not* give is that our inner routine attains that inner maximum. The per-user 1-D maximizations are exhaustive on a grid followed by refinement, so each attains its own grid optimum exactly; the step without a guarantee is the mean-field coupling between users, for which we have no contraction proof and therefore report empirical evidence only: agreement with a brute-force joint search at $N=2$, convergence within six iterations, and insensitivity to the initialization. We state the boundary between the proved and the measured explicitly rather than letting the word *exact* cover both.

B. Mean-Field Block-Coordinate Inner Solver

Inner couplings inside (I) are of two kinds: through the shared β_A (in every $P_{\text{corr}}^A, P_{\text{err}}^A$), and through the exclusion field $\phi_i = \prod_{j \neq i} (1 - P_{\text{corr}}^j)$ in (10). We treat ϕ as a field

Algorithm 1 Decomposed solver for **(P)**

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1: Input: cluster state  $\mathbf{s}$ , grids  $\mathcal{M}_\mu, \mathcal{M}_\chi$ , tolerances  $\epsilon_\phi, \epsilon_R$ 
2: Output:  $\mathbf{p}^*, V^*$ 
3:  $V^* \leftarrow -\infty$ 
4: for all  $(\mu, \chi) \in \mathcal{M}_\mu \times \mathcal{M}_\chi$  do
5:   Init  $\phi^{(0)} \leftarrow \mathbf{1}$ ,  $\beta_A^{(0)} \leftarrow \beta_{10}$ 
6:   for  $k = 0, 1, \dots$  until  $\|\phi^{(k+1)} - \phi^{(k)}\| < \epsilon_\phi$  do
7:     for  $i = 1, \dots, N$  do
8:        $\beta_i^* \leftarrow \arg \max_\beta R_s^i(\beta; \mu, \chi, \beta_A^{(k)}, \phi_i^{(k)}) \sigma_i(\cdot)$  sub-
       subject to per-pair constraints {1-D search}
9:     end for
10:     $\beta_A^{(k+1)} \leftarrow \arg \max_{\beta_A} \sum_i R_s^i \sigma_i$  holding  $\{\beta_i^*\}$  {1-D
    search, block coordinate}
11:     $\phi_i^{(k+1)} \leftarrow \prod_{j \neq i} (1 - P_{\text{corr}}^j(\beta_j^*))$  {field refresh}
12:  end for
13:   $V \leftarrow \sum_i R_s^i \sigma_i$  at current  $(\mu, \chi, \beta_A, \{\beta_i\})$ 
14:  if  $V > V^*$  then
15:     $V^* \leftarrow V$ ,  $\mathbf{p}^* \leftarrow (\mu, \beta_A, \{\beta_i\}, \chi)$ 
16:  end if
17: end for

```

and iterate the procedure listed in Algorithm 1 and depicted in Fig. 2:

The flow is depicted in Fig. 2.

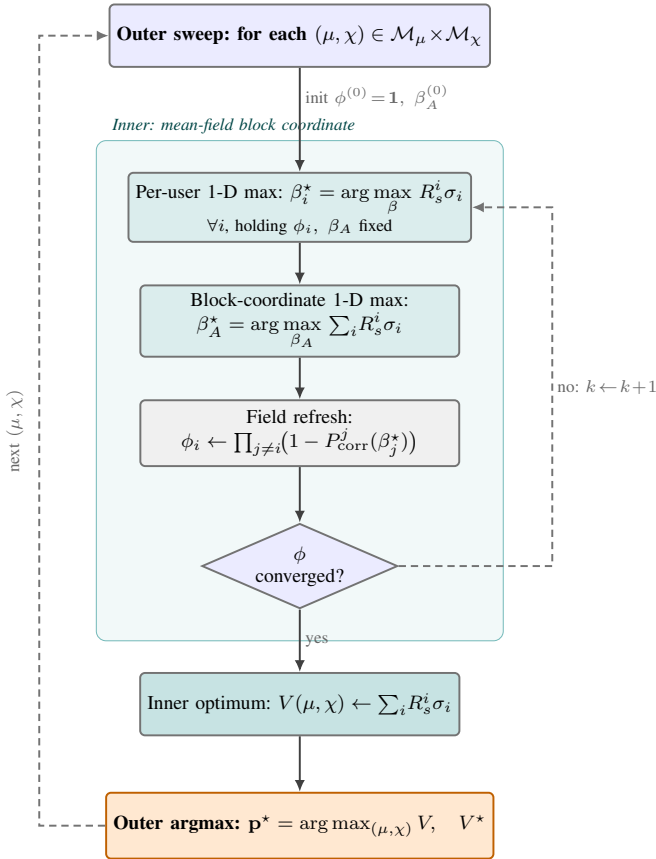


Fig. 2. Decomposed solver. The outer grid sweeps (μ, χ) ; the inner mean-field block-coordinate loop alternates per-user 1-D maximizations over β_i (held the field ϕ_i and the shared β_A fixed), then refreshes β_A by 1-D maximization, then refreshes the field ϕ_i from the new $\{P_{\text{corr}}^j\}$. The fixed point is the inner optimum $V(\mu, \chi)$; the outer argmax is $\mathbf{p}^*$.

C. Correctness Checks

1) *Symmetric-cluster reduction:* For a symmetric cluster ($s_i = s_*$ $\forall i$, $\beta_i = \beta_B$ $\forall i$), the field collapses to $\phi = (1 - P_{\text{corr}}^B)^{N-1}$ and the excluded sift (10) reduces to $P_{\text{corr}}^A P_{\text{corr}}^B [1 - (1 - P_{\text{corr}}^B)^{N-1}]$, recovering [11, Eq. 24].

2) *Brute-force agreement:* On a two-user cluster the decomposition is verified against a brute-force joint grid over $(\beta_A, \beta_1, \beta_2)$ at a fixed (μ, χ) node: the decomposed and brute-force optima agree, with a ratio of 1.000 to four decimals (Sec. VI-B). The closed-form exclusion field (10) is separately checked against a direct subset-enumeration inclusion-exclusion sum for $N \in \{1, 2, 4, 6\}$ (agreement to 10^{-12}).

D. Convergence and Complexity

The solver uses $|\mathcal{M}_\mu| = 12$ outer nodes uniformly spaced on $[0.4, 0.95]$ and $|\mathcal{M}_\chi| = 5$ on $[0, 1]$ in the headline runs. The inner field iteration converges in ≤ 6 steps from any cold start at the working tolerances $\epsilon_\phi = 10^{-3}$ and $\epsilon_R = 10^{-4}$; the iterate reached from any single initialisation converges, but the fixed point is *not* unique: restarting the inner iteration from one common- β value at a time, all 12 of 12 tested cluster states reach different fixed points depending on the start, with β_A^* differing by up to 1.31. Multi-start is therefore not a convenience but a requirement, and the earlier draft of this paper overstated the situation by reporting uniqueness; what is unique is the best-of-four solution the solver returns, not the fixed point itself. A formal characterisation of when the map is contractive is left for future work. One full outer sweep (all $12 \times 5 = 60$ outer nodes, each with up to 6 inner iterations and Gauss-Hermite quadrature at $n=20$) takes approximately 4 s on a single CPU core, so a full pass sampled at $\Delta t = 50$ s (~ 50 timesteps $\times 4$ s) solves in roughly 200 s. Multi-start is enabled for the inner 1-D maximizations, because the feasibility boundary creates several disconnected operating windows in β for certain (μ, χ) choices; we restart the inner iteration from four fixed common- β values $\{0.6, 1.2, 1.8, 2.4\}$ and retain the best feasible solution, ranked by feasible-pair count and then by total rate. The starts are deterministic, so the solver is reproducible without a seed.

E. Why This Decomposition Works

Three properties make the outer / inner split practical. First, the outer block is genuinely two-dimensional: μ and χ are the only variables that couple all N pairs through threats (URA via μ) and through the exclusion field (via χ). Second, given the outer block, the inner subproblem (I) admits a mean-field decoupling in the user index i : with ϕ frozen, each β_i^* is determined by a 1-D maximization that does not see the other users, and the field is refreshed at the next outer iteration. Third, β_A couples back through Alice's link only, so a single 1-D block-coordinate update suffices. The whole procedure is therefore exponentially cheaper than the naive joint $(N+1)$ -dimensional grid search, while preserving exactness in the symmetric special case (Sec. VI-B).

V. TWO-STAGE KAN CONTROLLER

A. Variable- N Architecture

The cluster controller must predict (μ, β_A, χ) at the cluster level and $\{\beta_i\}_{i=1}^N$ at the user level, where N may vary across deployments. We adopt a two-stage architecture (Fig. 3) that handles arbitrary N by parameter sharing:

- *Global head* $\mathcal{H}_g$: input $\mathbf{s}_g = (s_A, N, \bar{s}_B, \min_i s_i, d_E)$; output $(\hat{\mu}, \hat{\chi}, \hat{\beta}_A)$.
- *Per-user shared head* $\mathcal{H}_u$: input $(s_i, s_A, \hat{\mu}, \hat{\chi})$; output $\hat{\beta}_i$. The same head is invoked once per user i .

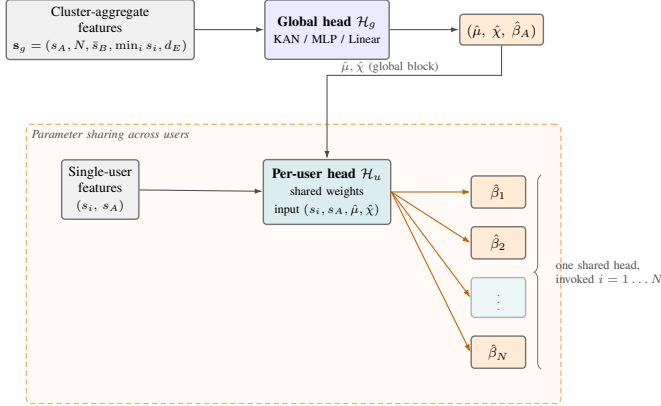


Fig. 3. Two-stage controller. The global head consumes cluster-aggregate features and predicts (μ, χ, β_A) ; the per-user head consumes single-user features plus the predicted global block and predicts β_i . Parameter sharing across users renders the controller dimension independent of N .

B. Hypothesis Classes

We compare three deployable model families for each head on the multi-axis trade-off, plus a non-parametric lookup reference (KNN) used only in the single-link comparison:

1) *Kolmogorov–Arnold Network (KAN)*: KAN [30] replaces the fixed pointwise nonlinearities of an MLP with *learnable* univariate functions on every edge, represented as a B-spline plus a residual SiLU basis. The motivation traces back to the Kolmogorov–Arnold representation theorem, which states that any multivariate continuous function on a bounded domain decomposes as a sum of univariate compositions; KAN parameterizes those univariate components as splines and fits them by gradient descent. We use the `pykan` [31] implementation at width $[d_{\text{in}}, 3, d_{\text{out}}]$, B-spline order 3, and grid size 5. The training pipeline has three stages: (i) standard regression to fit the splines; (ii) pruning of edges whose contribution falls below a threshold; (iii) symbolic candidate fitting from a library of polynomial, sin, exp, and tanh functions, followed by a joint refit of the network with the chosen symbolic substitution in place. The refit step is critical: skipping it collapses the apparent closed-form to a constant predictor, which was an early pitfall we corrected. The closed-form rule for β in Sec. VI-F is then read off the trained splines using the methodology of [36].

We state at the outset what the controlled KAN-versus-MLP literature leads us to expect, because our own results land exactly where it predicts. Comparing the two families

at matched parameter counts and FLOPs across machine learning, vision, audio and language tasks, [40] reports that MLPs generally outperform KANs *except* on symbolic formula representation, and [41] finds the parameter overhead of KANs a liability in data-scarce regimes. Our setting is data-scarce by construction, since each training pair costs one call to the decomposed oracle. We therefore do not expect, and do not find, a decisive regression advantage for the KAN over a regularized MLP on the one non-degenerate output; what we do find is the symbolic extraction that [40] identifies as the KAN’s remaining advantage, together with a parameter count 43% of the MLP’s. Reading our result as a specific instance of that pattern is more defensible than reading it as a general win, and we present it that way.

2) *Regularized Multi-Layer Perceptron (MLP)*: Two hidden layers of width 32, ReLU, weight decay 10^{-3} , early stopping on a held-out 20% split. Adam at 10^{-2} , 1500 epochs. This is a deliberately strong baseline; an unregularized MLP overfits the small dataset and produces unfair comparisons (we verified this experimentally before fixing the protocol; see Sec. VI-F).

3) *Linear Residual*: A linear least-squares fit of all outputs against all features; serves as the residual base for the KAN/MLP heads.

4) *k-Nearest-Neighbor Lookup (KNN)*: A non-parametric table baseline that predicts each output as the distance-weighted average of its $k=5$ nearest neighbors (Euclidean, standardized features) in the training set. It is included in the single-link comparison as a memorization reference: it interpolates the oracle well in regression but, being a lookup table, offers neither parameter economy nor an interpretable rule.

C. Residual Learning

Direct regression on the cluster-controller dataset proved fragile. Regression MAE remained small (between 10^{-3} and 10^{-2}), yet closed-loop key retention collapsed to zero, because the oracle sat on a sharp 4-D feasibility boundary in $(\mu, \chi, \beta_A, \beta_i)$ space, and multi-directional errors from KAN and MLP pushed predictions across that boundary. The linear baseline, ironically, attained 0.56 key retention by virtue of its smoothing bias toward the cluster mean.

We therefore adopt a *residual learning* strategy. The linear baseline $\mathcal{H}_{\text{lin}}$ is fit first by ordinary least squares; KAN or MLP then regresses on the residual $y - \mathcal{H}_{\text{lin}}(x)$, and the deployed predictor is the sum $\mathcal{H}_{\text{lin}}(x) + \mathcal{H}_{\text{resid}}(x)$. The output layer of the residual network is zero-initialized, so the initial prediction is exactly the linear one; training then carves out a corrective signal only where the linear approximation hurts feasibility. Empirically this lifts feasibility for every controller (Sec. VI-G). We note that what an earlier draft of this paper called a *boundary collapse* of KAN and MLP was not a property of the feasibility boundary at all but the effect of two software faults, identified and corrected in Sec. VI-G; the margin recipe is still useful, but it is not rescuing the controllers from a modelling pathology without sacrificing the linear baseline’s smoothing bias. The same idea has been used in classical signal processing under the name *boosted*

regression, and in physical-layer deep learning under *residual model-based augmentation* [32], [33].

D. β Safety-Margin Deployment Recipe

Even after training, predicted $\hat{\beta}$ values that fall just below the BSA $m^{\min}$ floor forfeit the entire pair-step. We deploy $\beta_{\text{deploy}} = \hat{\beta} + \delta$ with a single scalar δ^* chosen by 1-D grid sweep on a held-out fold. The curve $\delta \mapsto \text{KeyRetention}$ is unimodal across all controllers, with $\delta^* \in [0.03, 0.10]$, lifting feasibility by $1.1\text{--}1.5\times$. The recipe is model-agnostic and adds zero inference cost.

E. Five-Axis Evaluation Protocol

Each controller is scored on five axes:

- M_1 : mean absolute error per output, $\text{mean} \pm \text{std}$ over five seeds.
- M_2 : closed-loop secret-key retention: ratio of achieved key (controller predictions plugged into the analytic environment) to oracle key (decomposed solver), feasible-gated.
- M_3 : parameter count: total trainable parameters (heads combined).
- M_4 : inference latency on a single CPU core, μs per prediction.
- M_5 : interpretability: whether a closed-form rule with $R^2 \geq 0.95$ can be extracted (boolean per output).

We deliberately report all five rather than collapsing to a single “score,” because the controllers occupy a Pareto front; the right model for deployment depends on the constraint mix. Interpretability is decisive for safety-critical satellite firmware, while raw key retention dominates for an airborne ground link, and parameter count matters for an embedded payload.

VI. NUMERICAL RESULTS

A. Setup

System parameters follow [11, Table I] and are reproduced in Table II for self-containedness. The peak optical power $P=32$ dBm at the GEO source, combined with the EDFA gain $G_a=40$ dB at each LEO relay, gives a link-budget-limited mean SNR amplitude γ_0 that ranges from approximately 0.06 at 60° zenith to 1.8 near overhead (i.e. the geometry alone induces a $30\times$ variation along a single pass, the central motivation for adaptive control). The Hufnagel-Valley parameter $w=21$ and ground-level $C_n^2(0)=10^{-15}$ correspond to clear-night turbulence at coastal Vietnamese latitudes, calibrated against the daytime profile in [19].

Headline runs are reported for $N=3$ heterogeneous Bob users at zenith offsets $\{0, 8, 16\}^\circ$ from the cluster centroid, with Eve standoff distances $\{26, 18, 35\}$ m for B_1, B_2, B_3 respectively. The asymmetry deliberately captures both close adversary placements (Bob₂ at 18 m, tightening the URA constraint) and far placements (Bob₃ at 35 m, slackening it), so that the per-user β_i^* must differ in any nontrivial optimum, justifying the per-user threshold variable. Two orbit sources are evaluated:

- *Analytic*. Ten staggered LEOs (closest approach every 250 s, $\theta_{\min} \in \{0^\circ, \dots, 4^\circ\}$, $H_L=550$ km, $\theta_{\min}^{\text{vis}}=30^\circ$) hand-picked to span the full $0\text{--}60^\circ$ zenith range.
- *TLE Walker*. n synthetic Starlink-shell satellites in $P=n/10$ orbital planes (RAAN spread 360°) with $S=10$ satellites per plane and a Walker delta inter-plane phase offset matching the operational SpaceX deployment pattern [23], [24]. Orbits are SGP4-propagated [21], [22] via the `skyfield` and `sgp4` Python libraries from epoch 2026-05-23T12:00:00 UTC over the PTIT/Hanoi ground station (21.005°N , 105.843°E). The bundled `starlink_synth.tle` round-trips through `sgp4.exporter.export_tle` for offline reproducibility; real Celestrak [42] dumps are accepted via the same code path with a single command-line flag.

B. Decomposed Solver as Oracle

On a two-user cluster the decomposed solver value $V_{\text{dec}}(\mu, \chi)$ is compared against the brute-force joint grid value $V_{\text{bf}}(\mu, \chi)$ over $(\beta_A, \beta_1, \beta_2)$ at a fixed outer node: the ratio is $V_{\text{dec}}/V_{\text{bf}} = 1.000$ to four decimals, and the inner field iteration converges in ≤ 6 steps from a cold start. As an independent check on the multi-user structure, the closed-form exclusion field (10) matches a direct subset-enumeration inclusion–exclusion sum to 10^{-12} for $N \in \{1, 2, 4, 6\}$. We henceforth treat the decomposed solver as the oracle for the supervised controller dataset. A brute-force optimality check for asymmetric $N > 2$ clusters is left to future work; the present evidence is structural exactness of the split (Sec. IV) plus the $N=2$ numerical agreement.

C. Headline: Adaptive vs Best-Static

Table III reports the headline adaptive-vs-static comparison on three orbit settings (single-link analytic, cluster analytic, cluster TLE-Walker $n=500$ long window). The corresponding per-step traces are shown in Fig. 4.

Three observations follow from Table III and Fig. 4.

Single-link motivates the controller. The $2.07\times$ key gain on the single-link analytic baseline establishes that even in the absence of cluster coupling, a time-varying $(\gamma_0, \sigma_X, w_{\text{eq}})$ trajectory benefits substantially from per-step parameter selection. The factor 3.07 on secure time is larger than the factor on key because the adaptive controller does not merely *improve* the rate, it *extends the operating window* into geometries that static QKD must concede.

Cluster results are tighter but qualitatively similar. The cluster analytic gain ($1.64\times$ key, $2.5\times$ secure pair-time) sits below the single-link gain, because pair feasibility requires both endpoints to clear the per-link SNR threshold simultaneously, restricting the operating window to near-overhead geometries. The asymmetric cluster of Sec. VI-A amplifies this effect: B_2 near a close Eve binds the URA constraint earliest, while B_3 far from Eve binds it latest, so the static $\mathbf{p}$ must compromise across users.

Realistic-orbit operation requires adaptation. The strongest gain ($2.10\times$ key, $3.6\times$ secure pair-time) appears on the most realistic setting, the long-window TLE Walker shell. Crucially,

TABLE II
SYSTEM PARAMETERS (AFTER [11, TABLE I]). TWO-PANE LAYOUT FOR COMPACTNESS.

Parameter	Symbol	Value	Parameter	Symbol	Value
GEO altitude	H_C	35 793 km	GEO beam divergence	θ_C	10 μ rad
LEO altitude	H_L	550 km	LEO beam divergence	θ_L	50 μ rad
Wavelength	λ	1550 nm	LEO aperture radius	a_L	10 cm
Bit rate	R_b	1 Gb/s	User aperture radius	a_U	5 cm
Peak optical power	P	32 dBm (1.585 W)	Visibility	V	30 km
PIN responsivity	R_e	0.9 A/W	Cn ² at ground	$C_n^2(0)$	$10^{-15} \text{ m}^{-2/3}$
EDFA gain at LEO	G_a	40 dB	HV wind parameter	w	21
ASE inversion factor	n_{sp}	5.0	Min elevation	$E_{\min}$	30°
Optical bandwidth	B_0	250 GHz	Eve offset	d_E	26 m
Noise bandwidth	Δf	0.5 GHz	Sift / QBER thresholds	p_s, q	$10^{-3}, 10^{-3}$
Noise figure (linear)	F_n	2.0	Eve-error / BSA min	$p_e, m^{\min}$	0.1, 0.5%
Sun spectral irradiance	$\mathcal{R}_{\odot}$	0.1 W/(cm ² · μ m)	Gauss-Hermite order	n	20

TABLE III
ADAPTIVE VS BEST-STATIC HEADLINE. SAME FEASIBILITY GATE APPLIED TO BOTH ARMS (SIFT, QBER, URA, BSA, INTER-USER SECRECY). n IS THE CONSTELLATION SIZE; T THE HORIZON.

Setting	n	T [s]	served	N	static (μ, β, χ)	static secure	adaptive secure	key gain
Single-link, analytic	10	3000	86 / 600	1	(0.50, 3.00, −)	28 / 600 (4.7%)	86 / 600 (14.3%)	2.07 ×
Cluster, analytic	10	3000	46 / 60	3	(0.95, 1.50, 1.00)	4 / 138 (2.9%)	10 / 138 (7.2%)	1.64 ×
Cluster, TLE Walker (long)	500	5400	67 / 108	3	(0.95, 1.50, 1.00)	5 / 201 (2.5%)	18 / 201 (9.0%)	2.10 ×
Cluster, TLE Walker (sparse)	200	3000	29 / 60	3	(0.40, 0.50, 0.00)	0 / 87	2 / 87 (2.3%)	n/a [†]

Secure-time gains (adaptive / static): $3.07\times, 2.50\times, 3.60\times$ (rows one to three). [†]On this epoch the static baseline achieves zero secure pair-steps, so no ratio is defined and we report the counts. Every row here is a single TLE epoch; Section VI-D repeats the two TLE rows over 10 epochs and reports the spread.

on the sparser $n=200$ shell over the shorter 50-minute window, the static baseline collapses to *zero* secure pair-steps on this epoch. Over 10 epochs it reaches zero on 5 of them and never exceeds 1, while adaptation never falls below 2 (Sec. VI-D). When the constellation density falls below the observability budget over the chosen station, a single $\mathbf{p}$ buys at most one secure pair-step across the full pass, and per-step adaptation is what keeps the link usable at all. That is the qualitative claim distinguishing adaptive QKD from optimization: the difference is between a link that carries key and one that essentially does not, rather than between high and low key rate.

D. One epoch is not a measurement: repeating the TLE rows

Each row of Table III is one TLE epoch over one ground station. A ratio computed from a handful of events on a single realization is a sample, not a measurement, so we repeated both TLE settings over 10 consecutive daily epochs at the same station, re-tuning the best-static parameters on each epoch so that the comparison stays conservative. The epoch reported in Table III is the first of the ten and reproduces its published counts exactly, which is what licenses reading the rest.

On the 500-satellite shell the static baseline yields 1 to 5 secure pair-steps per epoch (mean 3.0) against 15 to 21 for adaptation (mean 18.5), pooling to $185/30 = 6.2\times$. The single-epoch figure of $3.60\times$ in Table III is therefore *conservative*: the epoch we reported is one on which the static baseline does its best over the ten, 5 pair-steps, and the pooled advantage is larger than the number the abstract carries. We have left the abstract at the single-epoch figure rather than raise it.

On the 200-satellite shell the correction runs the other way, and against us. The static baseline is not always shut out: it reaches zero on 5 of 10 epochs and one secure pair-step on the rest, mean 0.5, against 2 to 6 for adaptation. The statement that no static parameter choice is secure at all on that shell holds for the epoch we first measured and not for the shell, and we have corrected it wherever it appeared. The defensible form is the pooled ratio, 42 against 5 secure pair-steps or $8.4\times$, together with the observation that the static arm never exceeds one pair-step on any epoch while adaptation never falls below 2.

E. TLE Walker Coverage

We calibrate the synthetic Walker shell against the analytic baseline by sweeping shell size over PTIT/Hanoi for a 2-hour window. Served-time coverage rises from 6.9% at $n=50$ to 49.7% at $n=200$ (just above the analytic-baseline 46.8%) and to 62.9% at $n=500$, matching the order-of-magnitude density of the operational Starlink shell over Vietnamese latitudes.

F. Single-Link Controller: Five-Axis Trade-Off

Table IV reports the controller comparison on the single-link Phase-3 dataset (488 states, 61 zenith $\times$ 8 Eve-distance, 198/50 feasible split, 5 seeds).

a) *Closed-form β rule (extracted from KAN)*: The trained KAN predicts β^* with full-network test $R^2 = 1.00 \pm 0.00$ across five seeds (column R_{net}^2 of Table IV). The regularized MLP reaches the same 1.00 ± 0.00 there, so the network fit does not separate the two hypothesis classes; what separates

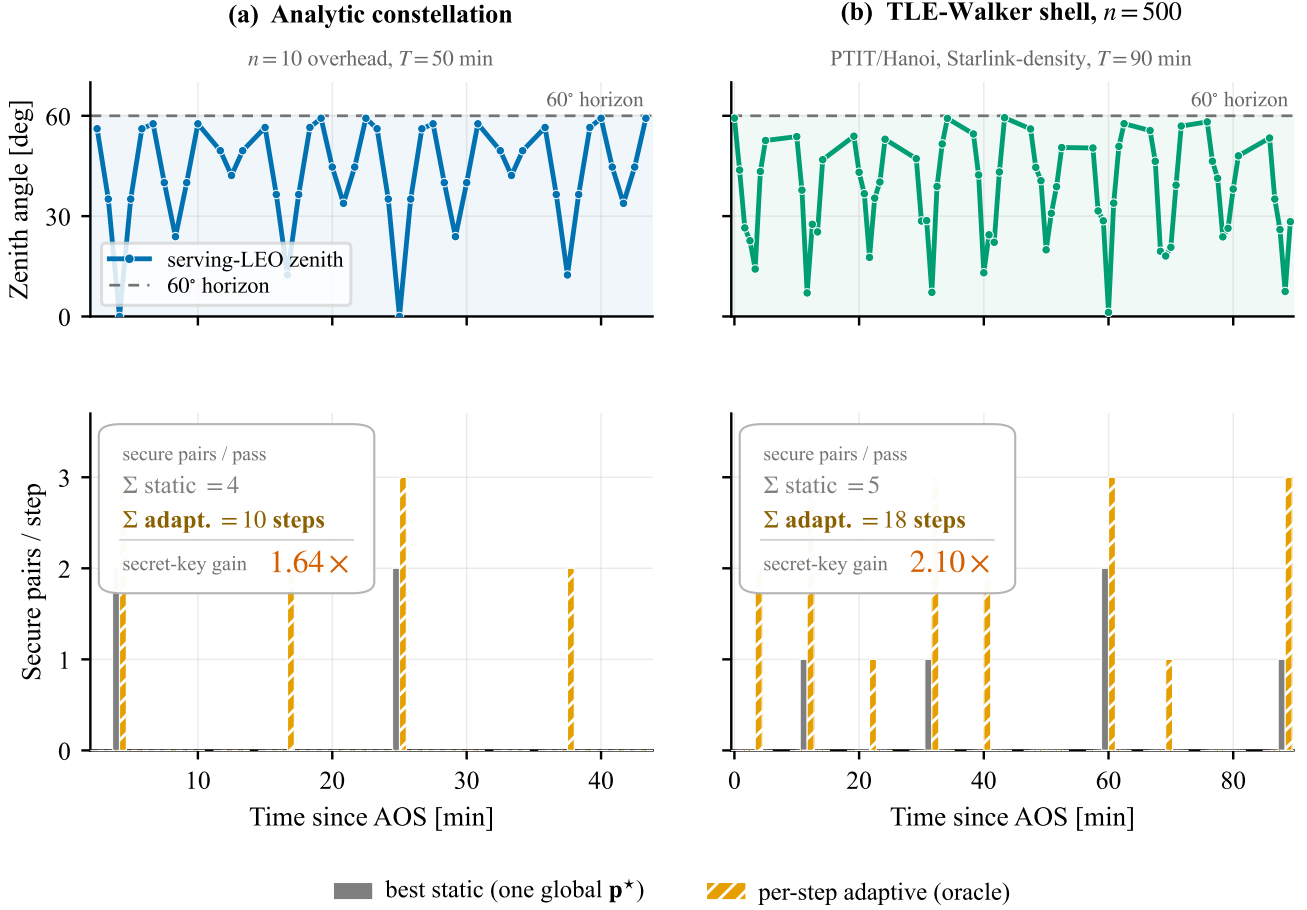


Fig. 4. Cluster QKD: adaptive vs best-static over a LEO pass. Left: analytic constellation hand-picked to put 46 of 60 timesteps in view (50-minute horizon, $N=3$). Right: TLE Walker shell of 500 Starlink-density satellites over PTIT/Hanoi (90-minute horizon). Top row: zenith trajectory of the serving LEO. Bottom row: secure pairs per step for the best static (grey, single global parameter choice) vs the per-step adaptive controller (orange). Adaptation lifts secure pair-time by $2.5\times$ on the analytic constellation and by $3.6\times$ ($5 \rightarrow 18$ pair-steps) on the realistic TLE shell. On a sparser shell ($n=200$, 50-minute horizon, last row of Table III) the static baseline reaches zero secure pair-steps on this epoch; Sec. VI-D gives the spread over 10 epochs.

TABLE IV

SINGLE-LINK CONTROLLER, FIVE-AXIS TRADE-OFF (MEAN $\pm$ STD OVER 5 SEEDS, 198/50 TRAIN/TEST FEASIBLE STATES). LEADING 0. DROPPED ON THE MAE AND KEYRET COLUMNS. R_{NET}^2 IS THE *full-network* TEST FIT ON β^* ; IT IS *not* THE POST-EXTRACTION SYMBOLIC FIT, WHICH IS WHAT THE M_5 BAR OF 0.95 IS APPLIED TO AND WHICH REACHES $R^2 \approx 0.97$ (SEC. VI-F0A). THE TWO ARE DISTINCT, AND ONLY THE KAN ADMITS THE EXTRACTION, WHICH THE “SYM.” COLUMN RECORDS. t_{INF} IS INFERENCE LATENCY PER PREDICTION ON ONE CPU CORE; THE KAN IS THE SLOWEST OF THE FOUR.

Model	MAE μ^*	MAE β^*	R_{net}^2	KeyRet	$t_{\text{inf}}[\mu\text{s}]$	#par	Sym.
KAN	.004 $\pm$.001	.011 $\pm$.002	1.00 $\pm$.00	.716$\pm$.123	116.7	392	yes
MLP	.008 $\pm$.001	.031 $\pm$.005	1.00 $\pm$.00	.683 $\pm$.086	5.0	1 282	no
KNN	.004 $\pm$.000	.045 $\pm$.000	0.99 $\pm$.00	.562 $\pm$.000	25.3	1 188	no
Linear	.023 $\pm$.000	.115 $\pm$.000	0.96 $\pm$.00	.614 $\pm$.000	0.1	10	no

them is whether a closed-form rule can be extracted from the fitted model at all, and only the KAN admits it. After pruning the KAN, running the symbolic candidate fit on a polynomial / sin / tanh library, and refitting jointly with the symbolic

substitution, a representative seed yields the closed-form rule

$$\begin{aligned} \beta^* \approx & 2.42 - 0.10 \gamma_0 - 0.03 \sigma_X + 0.23 w_{\text{eq}} \\ & + 1.24 \sin(0.50 \sigma_X + 5.27) \\ & - 0.30 \sin(0.48 w_{\text{eq}} + 2.04) \\ & - 0.17 \tanh(9.2 d_E + 8.62), \end{aligned} \quad (21)$$

as a function of the per-link features ($\gamma_0, \sigma_X, w_{\text{eq}}, d_E$), whose symbolic fit attains $R^2 \approx 0.97$ (above the 0.95 interpretability bar of M_5). The rule is dominated by the turbulence variance σ_X and the equivalent beam width w_{eq} through the spline-derived sin terms; the SNR scale γ_0 enters only weakly (linear coefficient -0.10), and the Eve standoff d_E enters through a saturating tanh that is flat over the operating range, consistent with d_E affecting feasibility more than the interior optimum. The point is that the optimal threshold admits a compact analytic surrogate at all, not that it is affine; we report the actual extracted form rather than an idealized linearization. The symbolic extraction for μ^* is performed but is not used in deployment, because the regression for μ^* saturates near its upper bound except for a sharp Eve-driven transition, and the symbolic R^2 is variable across seeds (cf. Sec. VII-B).

b) β safety-margin recipe: Fig. 5 shows that deploying $\beta_{\text{deploy}} = \hat{\beta} + \delta$ with a single scalar $\delta \in [0.03, 0.10]$ lifts the closed-loop key retention of every controller by 1.1–1.5 $\times$, with a unimodal curve. The optimum is model-dependent (KAN: $\delta^* \approx 0.03$, KNN: $\delta^* \approx 0.07$) and the recipe is zero-cost at inference. The KNN baseline gains the most precisely because it is the most boundary-sensitive at $\delta=0$.

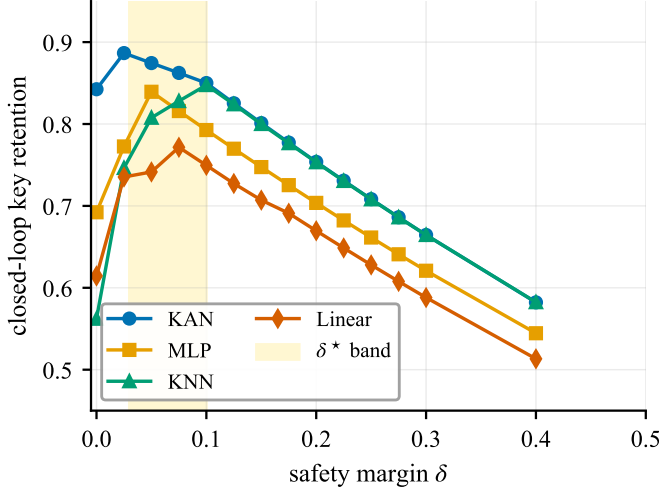


Fig. 5. β safety-margin deployment recipe: closed-loop key retention vs the scalar safety margin δ applied as $\beta_{\text{deploy}} = \hat{\beta} + \delta$. The unimodal shape and the model-agnostic peak in $[0.03, 0.10]$ make δ^* a single-knob deployment dial. KAN: 0.84 $\rightarrow$ 0.89; KNN: 0.56 $\rightarrow$ 0.83.

G. Cluster Controller: Two-Stage Residual Learning

Direct two-stage regression on the cluster controller dataset collapses for both KAN and MLP: regression MAE remains low ($\leq 10^{-2}$), but closed-loop key retention falls to zero because the oracle sits on a sharp 4-D feasibility boundary, and multi-directional errors push predictions outside. The linear baseline alone, by virtue of its smoothing bias, attains 0.56 key retention. The residual-learning rescue, in which KAN and MLP are trained on $y - \mathcal{H}_{\text{lin}}(x)$ with a zero-initialized output layer, restores both nonlinear models (Table V).

1) *Three of the four outputs are degenerate*: The optimal modulation depth saturates at the physical ceiling of the modulation index. On a 501-point sweep to $\mu=0.999$ the constrained optimum lies at the upper boundary in 80% of sampled channel conditions, and widening the sampled geometry raises that to 68% boundary plus 19% infeasible rather than producing interior optima. Consequently μ^* , β_A^* and β_i^* are effectively constant across the dataset, and every row of Table V attains sub- 10^{-3} error on them, including a nine-parameter linear map. Only χ carries learnable signal.

2) *What two software faults cost, and what remains after*: Block (a) of Table V reproduces the originally submitted configuration. Two faults were corrupting it. The feature standardiser guarded against an exactly-zero standard deviation, but the constant μ input has standard deviation 7.99×10^{-15} , so the guard was bypassed and floating-point noise was amplified by 1.25×10^{14} ; predicted thresholds moved by 0.35 in response to last-bit input differences, and the feasible

TABLE V
CLUSTER TWO-STAGE CONTROLLER, 20-SEED EVALUATION ON A SINGLE MACHINE (2 $\times$ RTX 4090, PYTORCH 2.6). *Div.* IS THE NUMBER OF NON-FINITE PREDICTED CELLS, REPORTED BECAUSE A DIVERGED FIT MUST NOT BE READ AS A MEASUREMENT. MAE μ , MAE β_A AND MAE β_i ARE BELOW 10^{-3} FOR EVERY ROW, BECAUSE THOSE THREE TARGETS ARE CONSTANT IN THIS REGIME (SEC. VI-G1); χ IS THE ONLY OUTPUT WITH SIGNAL. ROWS (a) REPRODUCE THE ORIGINALLY SUBMITTED CONFIGURATION AND ARE SHOWN ONLY TO QUANTIFY WHAT TWO SOFTWARE FAULTS COST.

Model	MAE χ	KeyRet	Feas.	Div.	#par
(a) as originally submitted (both faults present)					
KAN+resid	.042 $\pm$.034	.600 $\pm$.452	94/158	6 320	1 239
MLP+resid	.082 $\pm$.001	.230 $\pm$.395	36/158	0	2 823
Linear	.083 $\pm$.000	.626 $\pm$.000	97/158	0	35
(b) standardisation fault fixed only					
KAN+resid	.042 $\pm$.034	1.005 $\pm$.004	158/158	6 320	1 239
MLP+resid	.082 $\pm$.001	1.009 $\pm$.001	158/158	0	2 823
Linear	.083 $\pm$.000	1.009 $\pm$.000	158/158	0	35
(c) both faults fixed (the configuration we report)					
KAN+resid	.010$\pm$.001	1.001 $\pm$.000	158/158	0	589
MLP+resid	.018 $\pm$.022	1.002 $\pm$.002	158/158	0	1 386
Linear	.083 $\pm$.000	1.009 $\pm$.000	158/158	0	9

count became machine-dependent. Perturbing predictions by a controlled relative amount reproduces the effect causally: a relative perturbation of 10^{-15} moves the feasible count from 91 to 97, and 10^{-12} moves it to 158. Separately, fitting a network to a residual whose standard deviation is 1.74×10^{-16} makes the quasi-Newton step diverge on 20/20 seeds, yielding 6320 non-finite cells that a silent fallback replaced with the training median.

Block (b) isolates the first fault: feasibility recovers for every model, but the KAN still diverges and its accuracy on χ is unchanged. Block (c) fixes both. Three things follow. First, the closed-loop retention metric *saturates*: all three models, including the nine-parameter linear map, reach the oracle rate on every test configuration, so that metric no longer separates architectures and we do not use it to rank them. Second, on the one axis that does carry signal, KAN attains .010 $\pm$.001 against .018 $\pm$.022 for the MLP and .083 for the linear map, at 43% of the MLP's parameter count. Third, the comparison that matters operationally is stability rather than mean accuracy, and the mean gap does not survive a test.

3) *The accuracy advantage over the MLP is not established*: The seeds are shared between models, so a paired test is the right instrument, and we ran it. Over the 20 paired seeds an exact two-sided Wilcoxon signed-rank test on χ gives $p = 0.294$, with the KAN lower on 13 of 20 seeds. The unpaired comparison a reader would compute from Table V gives $t = 1.46$, and neither establishes a difference. We therefore do not claim an accuracy advantage over the regularized MLP, and have removed that reading from the abstract.

What produces the apparent gap is visible per seed. On 18 of 20 seeds the two are indistinguishable, 0.0102 for the KAN against 0.0103 for the MLP. On the remaining two the MLP returns the linear baseline's value, 0.083, meaning its residual

head contributed nothing on those seeds; the KAN does this on none. The entire difference in means, and the $26\times$ ratio of standard deviations, is that 2 of 20 against 0 of 20. That failure-rate difference is itself not established at this sample size either, Fisher's exact two-sided $p = 0.487$, and we report it as an observation rather than a result.

The comparison that does separate the two hypothesis classes on this axis is against the linear map, where the KAN is lower on 20 of 20 paired seeds, $p = 1.9 \times 10^{-6}$. That is the floor of the test at this sample size, so we report the effect: 0.0102 against 0.0829, an $8.1\times$ reduction in error.

4) *How many of the five axes separate the hypothesis classes:* It is worth doing this arithmetic out loud rather than letting the five-axis framing imply more than it carries. M_2 is dead: closed-loop retention saturates for all three models. M_1 is informative on one of the four controller outputs, the other three being constant, and on that one output it separates the KAN from the *linear map* but not from the regularized MLP (Sec. VI-G3). M_4 runs against the KAN, which is the slowest model at both scales, $116.7\mu\text{s}$ per prediction on the single-link head (Table IV) and $240\mu\text{s}$ on the cluster pair against 8.2 for the MLP.

Two axes are left, and they are the ones the contribution rests on: M_3 , where the KAN uses 43% of the MLP's parameters, and M_5 , where it is the only model of the three from which a closed-form rule can be extracted. Both are properties of the hypothesis class rather than outcomes of a noisy fit, which is why they survive while the accuracy axis does not. We frame the contribution on those two and say plainly that the KAN buys them at a cost in latency.

What the learned controller buys over the nine-parameter linear map is narrower still. On χ the linear map reaches .083 against the KAN's .010, an $8.3\times$ larger error, so the nonlinearity does real work on the one output that varies. That accuracy does not convert into key on this test set, because retention has saturated for the linear map too: within the operating envelope we sample, a nine-parameter affine rule already holds the oracle rate. The gap would begin to matter outside that envelope, which we have not measured and do not claim.

H. Robustness Across Conditions

We re-solved the oracle on 25 fresh clusters under each of sixteen conditions spanning four axes: atmospheric turbulence (wind 11 to 41 m/s in the Hufnagel–Valley profile), serving geometry (zenith bands from $0\text{--}4^\circ$ to $20\text{--}35^\circ$), cluster size ($N=2$ to 6), and eavesdropper standoff (10 to 120 m). Two observations survive the sweep and one of them surprised us.

The saturation is a property of the problem, not of one regime. The optimal modulation depth sits at its upper bound in 100% of clusters in every feasible condition, across every turbulence level, zenith band and cluster size we tried. The single exception is a close eavesdropper: at a standoff of 10–20 m the URA constraint binds and the optimum moves inside the range in just under half the cases (52% remain at the bound). The per-user threshold is likewise constant to within one grid cell everywhere except the $8\text{--}20^\circ$ zenith band.

Turbulence helps security, and calm air is the adversarial case. Two conditions produced no feasible cluster at all. The $20\text{--}35^\circ$ zenith band fails for the expected reason, a longer slant path. The low-turbulence case (11 m/s) fails for a reason worth stating: the beam-splitting attack is detected only through the loss it imposes on the sifting statistics, so detectability scales with the width of the fading distribution. Measuring at a fixed geometry, m_{BSA} rises monotonically with turbulence, from 0.0038 at 11 m/s, below the 0.005 threshold, through 0.0063, 0.0089 and 0.0105 at 21, 31 and 41 m/s. A clear, still night is therefore the worst case for this security criterion, not the best: the link is at its cleanest and a passive splitter is at its least visible. We did not anticipate this and report it as a design constraint rather than a result in our favour.

I. Single-Link Pass Trace

Fig. 6 traces the per-step (μ^*, β^*) chosen by the adaptive controller along a single LEO pass, together with the static baseline. The adaptive μ^* saturates near its upper bound whenever the URA constraint is slack (Eve far) and drops sharply when Eve approaches; β^* tracks the closed-form rule (21), varying primarily with the turbulence variance σ_X and the beam width as the geometry changes near zenith.

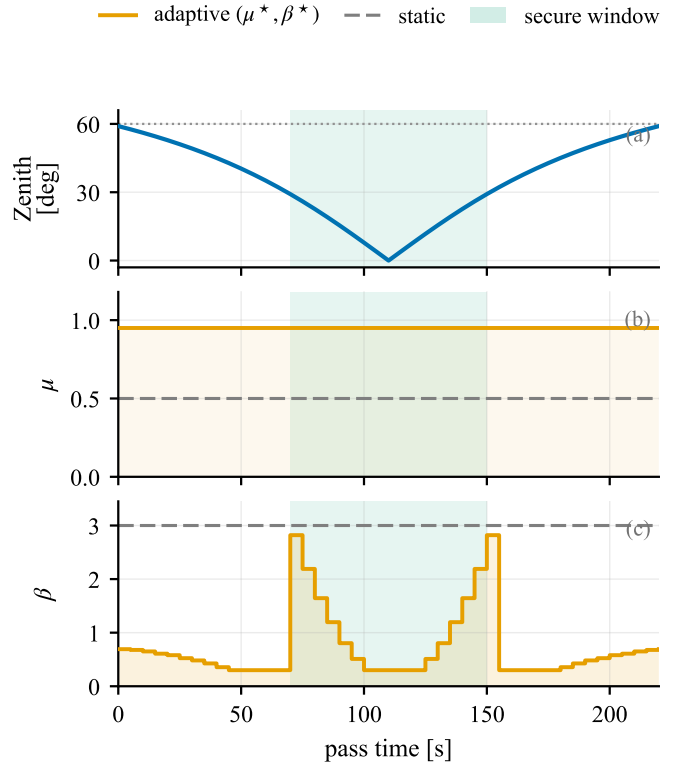


Fig. 6. Single-link adaptive control over a LEO pass. Top: zenith trajectory. Middle/bottom: adaptive μ^*, β^* (orange step) vs static (grey dashed). The β^* envelope is well-fit by the closed-form rule (21), justifying the KAN interpretability claim; the μ^* envelope saturates near μ_{max} except where Eve binds.

VII. DISCUSSION AND CONCLUSION

A. What Works, What Does Not

Adaptive parameter control buys a usable margin, and the defensible form of that claim is narrower than the one we made before Sec. VI-D measured it. Under Kepler-propagated orbits adaptation lifts the secure pair-time by $3.6\times$ on the reported epoch and $6.2\times$ pooled over 10 at a Starlink-density shell. On a sparser shell it raises the best static arm's *ceiling* of 1 secure pair-step to a *floor* of 2, $8.4\times$ pooled: necessary for any usable margin, but survival rather than operation, not the difference between operating and not. The decomposed solver's outer/inner split is exact and matches the brute-force joint optimum on a two-user cluster. The honest summary of the controller comparison is narrower than we first reported: closed-loop retention saturates for every architecture including a nine-parameter linear map, so the separation lives in accuracy on the single non-degenerate output, in the stability of that accuracy across seeds, and in parameter count and symbolic extractability, while inference latency runs against the KAN (Sec. VI-G4).

B. Limitations

Partial interpretability. The KAN extracts a clean closed-form rule for β^* but not for μ^* . The latter saturates near $\mu_{\max}$ except for a sharp Eve-driven transition that the symbolic candidate library (polynomial, exp, tanh) cannot fit with $R^2 \geq 0.95$. A wider library (sigmoid, ReLU surrogates) is a candidate fix; we report the regime honestly rather than overstating KAN's interpretability. The saturation is not a shortcoming of the symbolic extractor alone: μ^* genuinely has almost no variation to express.

Degenerate targets, and what they cost. Three of the four controller outputs are constant over the operating regime, because the optimal modulation depth saturates at the physical ceiling of the modulation index: on a 501-point sweep to $\mu=0.999$ the optimum lies at the upper boundary in 80% of sampled channel conditions, and widening the sampled geometry makes that fraction worse rather than better. A learned controller therefore has one axis worth learning, not four, and we report accuracy on that axis rather than averaging over three trivial ones. This degeneracy also carries a software cost, traced in Sec. VI-G2: both faults are now handled explicitly and the per-configuration divergence count is reported beside every row, so a diverged fit can never be read as a measurement.

Synthetic vs real TLE. The Walker shell reproduces the coverage statistics of the operational Starlink constellation but is not a snapshot of a specific real epoch. Loading a fresh Celestrak dump (`-tle PATH`) anchors the experiment to a specific UTC instant and is the figure that referees will check; the code path is identical.

Finite-key analysis. The reported rates are asymptotic; a finite-key analysis with composable security bounds, for example the tight bounds of Tomamichel *et al.* [43], is left to future work. We expect finite-key penalties to favor the adaptive controller further, since the penalty is heaviest where

the sift count is smallest, exactly the regime in which the static baseline collapses.

Single-station ground segment. The Walker headline is reported for a single ground station (PTIT/Hanoi). Multi-station handover (the obvious next-decade deployment scenario for satellite QKD, projected into trusted-node-free metropolitan QKD networks such as [44] and the long-term quantum-internet vision [45]) would soften the static-baseline collapse but is beyond the scope of this paper.

Scope of the eavesdropper model. URA and BSA are the two attacks for which this physical layer admits a closed-form detectability criterion, and they are modelled individually rather than jointly. Photon-number splitting, Trojan-horse probing, detector blinding and time-shift attacks on the dual-threshold receiver, side channels in classical post-processing, and collective or coherent attacks are all outside what we model. The security statement should be read accordingly.

Detection sensitivity is sharp. The beam-splitting criterion certifies detection only above a narrow threshold: the fraction of pairs for which detection can be certified falls from 100% at a 1.4% power split to zero at 1.15%, a transition occupying about a quarter of a percentage point. Against a stealthier splitter the constraint is simply not met and no key is emitted, which is safe but is also a real bound on what the design certifies.

Oracle dependence, and a saturated metric. The controller is trained by supervision from the decomposed solver, so it inherits whatever that solver gets wrong, and it cannot be better than its oracle by construction. It also cannot be better than a metric that has saturated: Sec. VI-G4 shows closed-loop retention no longer discriminates between architectures, and we rank on the axes that still do.

C. Future Work

Three directions are immediate. (i) Generalization sweep across visibility, turbulence outer-scale, Eve distance, and ground-station latitude, since the existing scripts are parametric in all four. (ii) Finite-key correction integrated into the per-step objective R_f ; this would tighten the secure-window estimate and likely favor the adaptive controller further, since finite-key penalties are heaviest at low sift. (iii) Multi-station fusion at the cluster head, treating the served-ground-segment trace as a partially-observed Markov decision process and re-deriving the controller as a constrained reinforcement learning agent. The current supervised pipeline already produces the oracle labels needed for an offline-RL warm start.

D. Conclusion

We have formulated, decomposed, solved, and learned a controller for the full multi-user joint optimization problem of GEO/multi-LEO/ N -user satellite QKD with SIM-BPSK/DT-DD. Validated against an SGP4-propagated Walker constellation matching Starlink density over a Hanoi ground station, the adaptive controller delivers a $2.10\times$ secret-key gain and a $3.6\times$ secure pair-time improvement, with a closed-form rule for the dual-threshold coefficient extracted from a Kolmogorov–Arnold Network. The static-parameter operating

point assumed in the preceding QKD literature is shown to be a special case that fails entirely under realistic orbit geometry, repositioning parameter adaptation from an optional optimization to a system-level requirement.

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